Matrices: Modelling

Questions 1–16

Question 1. In a sanctuary, there are three animal species: Wolves, Foxes, and Bears. On the 1st January 2022, the number of each animal species was counted.

a) Define variables for the number of each animal species on 1st January 2022.

b) If there are 100 more Foxes than Wolves in 2022, write an equation relating the number of Foxes and Wolves using the variables defined in a).

c) If the total population of all three species on 1st January 2022 was 1500, write an equation that relates the number of Wolves, Foxes and Bears to the total population.

On 1st January 2023, it was found that the total number of animals in the sanctuary had increased by 1%. For each species, it was found that the number of Wolves had increased by 3%, the number of Foxes had decreased by 5%, and the number of Bears had increased by 2%.

a) Write down separate expressions for the number of Wolves, Foxes, and Bears on 1st January 2023.

b) Write down an equation that represents the total population of animals on 1st January 2023.

Question 2. The food requirements for a café over three days are shown in the table below:

	Sandwiches	Salads	Smoothies
Day 1	25	15	10
Day 2	30	20	12
Day 3	28	18	14

The costs per item are: sandwiches cost £3.00 each, salads cost £4.50 each, and smoothies cost £2.50 each.

a) Construct a matrix equation to represent the total expenditure on each type of food for all three days.

b) Calculate the overall expenditure across all food items and days.

Question 3. The table shows the probabilities of a winter day being snowy in three Canadian cities.

	Dec	Jan	Feb	Mar
Toronto	0.25	0.30	0.28	0.20
Vancouver	0.15	0.12	0.10	0.08
Montreal	0.40	0.45	0.42	0.35

Show how you can use matrices to calculate the total number of snowy days expected in each city over the four months, given that it is not a leap year.

Question 4. The number of packages delivered by four drivers over three different months is shown in the table below.

	January	February	March
Driver A	120	150	140
Driver B	135	160	130
Driver C	110	140	125
Driver D	145	155	135

The company pays a delivery fee of £2.50 per package for January, £2.75 per package for February, and £3.00 per package for March.

a) Write a matrix to represent the table of deliveries and a vector to represent the delivery fee for each month.
b) Show how the matrices from a) can be used to calculate the total delivery fee accrued by each driver across all three months.
c) Write down the combined total delivery fee accrued by all four drivers across the three months.

Question 5. A warehouse's inventory consists of three electronic products: smartphones, laptops and tablets, with a total stock of 600 units. Originally, there were 200 more tablets than smartphones. After one year:

- The number of smartphones increased by 3%.
- The number of laptops increased by 5%.
- The number of tablets decreased by 8%.
- The total inventory decreased by 11.

Form and solve a matrix equation to determine the original stock levels of each product.

Question 6. Sidnie invested £5500 in shares of three companies: TechCorp, FinCorp, and RetailCo. Initially, she invested £1000 more in FinCorp than in TechCorp. After one year, the share values changed as follows:

- TechCorp's increased in value by 2.5%.
- FinCorp's decreased in value by 1.8%.
- RetailCo's increased in value by 3.2%.
- Overall, the total value of Sidnie's investments increased by £12.

Form a matrix equation to determine the original amount Sidnie invested in each company's shares.

Question 7. A colony of 1350 penguins includes three species: Adélie, Gentoo, and Chinstrap penguins. Originally, there were 210 more Gentoo penguins than Chinstrap penguins. After one year:

- The number of Adélie penguins decreased by 5%.
- The number of Gentoo penguins decreased by 10%.
- The number of Chinstrap penguins increased by 12%.
- The total number of penguins in the colony decreased by 48.

Form a matrix equation to determine the original number of Adélie, Gentoo, and Chinstrap penguins in the colony.

Question 8. A publishing company sells books across three genres: Fiction, Non-Fiction, and Children's books. It is known that:

- Each Fiction book sold generates £3 profit.
- Each Non-Fiction book sold generates £4 profit.
- Each Children's book sold generates £2.50 profit.

In 2023, due to market changes:

- Fiction sales dropped to 70% of the 2022 level.
- Non-Fiction sales increased by 30% compared to 2022.
- Children's book sales increased by 20% compared to 2022.

A total of 155,800 books were sold across all three genres. The total profit from all sales in 2022 was £443,000, and the total profit dropped by 5% in 2023. Form and solve a matrix equation to determine, to 3 significant figures, the number of books sold for each genre in 2022.

Question 9. It is given that P_n is the population of puffins on an island n years after the 1st January 2015. Write down, in context, what is meant by:

a) P_0

b) P_{10}

Question 10. It is given that $A_{n+1} = 0.51A_n + 0.32B_n$ where $A_0 = 2.5$ and $B_0 = 0.9$. Calculate A_1.

Question 11. It is given that $C_{n+2} = 0.63C_{n+1} + 0.28D_n$ where $C_1 = 2.7$ and $D_0 = 1.5$. Calculate C_2.

Question 12. A population of wolves is split into adults and pups. The survival rate of pups is 0.83. Each year, one quarter of the surviving pups transition to adulthood. Use the notation P_n, the pup population in year n, to form an equation for the pup population in year $n + 1$, justifying your reasoning based on survival and transitions to adulthood.

Question 13. In a fish population model, let F_n and L_n represent the number of adult fish and larvae respectively in year n. The population changes each year as follows:

- Adult fish survive at a rate of 0.88, and one fifth of the surviving larvae mature into adults each year.
- The larvae population survives at a rate of 0.75 each year.

Express the system of equations for F_{n+1} and L_{n+1} in matrix form.

Question 14. In a bee colony, the population is divided into workers and larvae. The system can be modelled by the following matrix equation:

$$\begin{pmatrix} W_{n+1} \\ L_{n+1} \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} W_n \\ L_n \end{pmatrix}$$

where W_n and L_n represent the number of worker bees and larvae respectively n years after 1st January 2024.

a) Describe in context what the top row of the matrix represents.
b) Describe in context what the bottom row of the matrix represents.

On 1st January 2024, the colony contained 4000 workers and 2000 larvae.

a) Use the given matrix to find the population of worker bees and larvae on 1st January 2025.
b) Assuming the rates in the model remain the same, apply the matrix a second time to find the population of worker bees and larvae on 1st January 2026.

Question 15. A wildlife reserve tracks the population of deer, which consists of yearlings and mature deer. Yearlings do not reproduce. In this study, the population at the start of each year is modelled using the transition matrix

$$\begin{pmatrix} Y_{n+1} \\ M_{n+1} \end{pmatrix} = \begin{pmatrix} k & 0.18 \\ 0.06 & 0.83 \end{pmatrix} \begin{pmatrix} Y_n \\ M_n \end{pmatrix}$$

where Y_n and M_n represent the number of yearlings and mature deer in year n after the study began, and k is a constant.

a) Explain what the constant k represents in the model.

At the start of the study, the total population was 72,000 deer, with 55,200 of these being mature deer. After one year, there were 46,824 mature deer and 15,816 yearlings.

a) Using the information provided, find an equation for Y_1 in terms of Y_0 and M_0, and hence determine the value of k to 3 significant figures.

b) Using the model and your value of k, calculate to 3 significant figures the predicted number of yearlings after two years.

c) Conservationists observed that the birth rate of yearlings declines after a certain threshold population is reached. Suggest how the matrix system could be modified to reflect this information.

Question 16. In a forest, the population of oak trees is made up of mature trees and saplings. Records have shown the following rates:

	Seed Production Rate	Survival Rate
Mature Trees	0.16	0.98
Saplings	0	0.74

It is assumed that 15% of the surviving saplings mature into mature trees each year. Let M_n and S_n denote, in thousands, the populations of mature trees and saplings, respectively, n years after 1st January 2020.

a) Write an equation for M_{n+1} in terms of M_n and S_n.

b) Write an equation for S_{n+1} in terms of M_n and S_n.

On 1st January 2020, the total oak tree population was estimated to be 2000 trees, with 1400 being mature trees.

a) Write down the initial population vector.

b) Write down the transition matrix T such that the population can be modelled by $\begin{pmatrix} M_{n+1} \\ S_{n+1} \end{pmatrix} = T \begin{pmatrix} M_n \\ S_n \end{pmatrix}$.

c) Use the model to find the total number of oak trees in the forest, to 3 significant figures: i) on 1st January 2019, ii) on 1st January 2024.

d) 100 mature oak trees are to be chopped down from the forest each year after 1st January 2020. Determine the total number of trees predicted to be in the forest on 1st January 2024 now.